



**JOHANNES KEPLER
UNIVERSITY LINZ**

UE MLPC 2026: DATA CLASSIFICATION TASK



2026-04-20

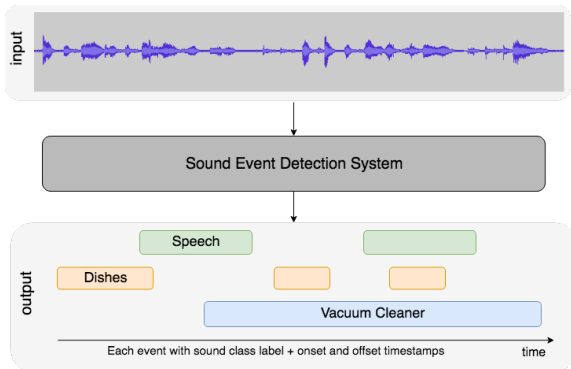
Institute of Computational Perception

Agenda

- Project Recap
- Project Schedule
- Data Exploration Task Recap
- Presentation of Selected Teams
- Data Classification: What to Look for?

Project Scenario

- *Kepler Intelligent Audio Labs (KIAL)* is an AI startup developing Sound Event Detection (SED) systems for **smart home environments**.



PROJECT SCHEDULE



Project Schedule (Meetings)

Date/Deadline

Meeting 1	Introduction, explain Data Collection	March 2 ✓
Meeting 2	Data Collection, explain Annotation	March 16 ✓
Meeting 3	Data Annotation, explain Exploration	March 23 ✓
Tutorial 1	(recording) NumPy, Pandas, Matplotlib	March 30 ✓
Meeting 4	Data Exploration, explain Classification	April 20 ◀
Tutorial 2	Data splits, Sklearn classifiers	April 27
Meeting 5	Classification, explain Challenge	June 1
Tutorial 3	PyTorch, advanced topics	June 1
Meeting 6	Final Presentations	June 29

Project Schedule (Tasks)

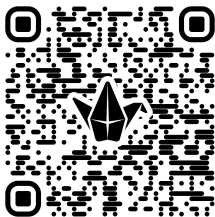
	Date/Deadline
Task 0 Form teams	March 16 ✓
Task 1 Data Collection	March 11 ✓
Task 2 Data Annotation	March 25 ✓
Task 3 Data Exploration Task	April 16 ✓
Task 4 Classification Task	May 28 ◀
Task 5 Challenge Task	June 25

DATA EXPLORATION TASK: RESULTS



Interactive Reflection and Group Presentations

- Have your reports open for reference
- Scan or click the QR code below¹:



- Use the platform to post your most interesting finding for each task, and your questions to us or to the presenter!

¹ <https://padlet.com/tarajadidi/mlpc26-exploration-i5650t5vxbykh3m9>

Task 1: Qualitative Analysis

■ Verify Annotations & Qualitative Analysis:

- (a) Verify the annotations assigned to you on Label Studio
- (b) Report mistakes and disagreements between annotators
- (c) Present one illustrative example

■ Selected team:

Algorithmic Learners

■ Remember: Use the provided link to post your questions



Task 2: Annotations

■ Annotations

- (a) Quantify annotator agreement: does it vary across classes and align with the metadata class tags?
- (b) Propose a strategy to convert annotations to Labels
- (c) Find patterns in the resulting labels

■ Selected team:

Embedded Engines



Task 3: Audio Features

■ Audio Features

- (a) Analyze metadata distributions
- (b) Calculate basic statistics of the audio feature values
- (c) Look into the correlation of the audio features (or groups of features)
- (d) **Bonus Task:** Create a visualization of the audio features

■ Selected team:

Incremental Networks



Task 4: Conclusions

■ Conclusions

- (a) Which biases were introduced during the data collection and annotation phase?
- (b) Is the dataset suitable for training SED models?
- (c) What are the societal impacts of making such a dataset publicly available?

■ Selected team:

Stochastic Pipelines



THE CLASSIFICATION TASK



Classification Task: Overview

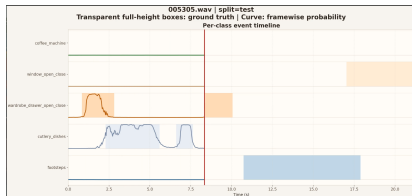
You have collected and explored the dataset, now it's time to build a model that can recognize sounds automatically.

In this task, you will go through the main steps of training a classifier:

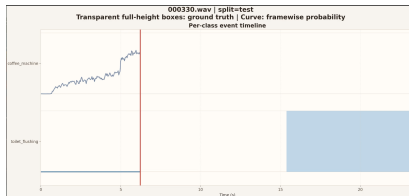
- Turn annotations into usable training labels
- Split the data into training and evaluation sets
- Choose how to measure performance
- Train and tune different classification models
- Analyze and interpret the model predictions

Goal: Understand the full machine learning pipeline and how your earlier data exploration impacts model performance.

Motivation: Classifier example



Click to open a classification example



Click to open an example of the model failing

What to Investigate 1/4

- **Dataset Preparation** (10 points): Prepare the labels, split the data and perform the necessary preprocessing steps for a machine learning task:
 - (a) **Label Aggregation:** Convert the annotations from multiple annotators into training labels for each time segment.
 - (b) **Data Split:** Describe how you split the data for model selection and performance evaluation. *You will need to train your model, tune hyperparameters, and estimate final performance.* Make sure to take the possibility of information leakage and the label distributions into account.
 - (c) **Preprocessing:** Explain any preprocessing techniques (eg. data normalization, feature selection) you used to prepare the data for all or certain classifiers.

What to Investigate 2/4

■ Evaluation (5 points):

- (a) Which evaluation criterion did you choose to compare hyperparameter settings and classifiers, and why?
- (b) What is the baseline performance? What could be the best possible performance?

What to Investigate 3/4

■ Experiments (12 points):

- (a) Choose at least two classifiers from different model classes. Look into the different hyperparameters each model has, and investigate and visualize how varying each of the values effects the model performance.
- (b) After selecting appropriate hyperparameters, compare the final performance estimate of the two chosen classifiers to the baseline performance established in 2.b.

What to Investigate 4/4

- **Case Study and Reflection** (10 points): Find two interesting audio files that were not used in training and analyze the models' predictions
 - (a) Use the spectrogram and the sequence of predictions to visualize the classifier output. How well does the classifier recognize the classes?
 - (b) What are typical failure cases? Which classes are detected reliably, and which ones are difficult to detect?

Data Classification Task: Report (37 points)

Compile your results into a short report.

- Cover **all** of previous aspects and the corresponding sub-questions
- use the [L^AT_EX template](#) that is available on Moodle
- include a “Disclosure of LLM and AI Tool Use”-section
- max. 6 pages
- max. 2000 words; rest figures, tables ...

Data Classification Task: Slides (3 points)

In addition to the detailed report, compile a short presentation

- Cover **one** of the four previous aspects and subquestions
- Your topic is determined by the first letter of your group name (see table below)
- max. 4 slides + 1 title slide

First Letter of Group Name	Topic
A-C	Case Study and Reflection
D-G	Experiments
H-Q	Dataset Preparation
R-Z	Evaluation

Data classification Task: Submission

- Submit your report and slide deck as two separate PDF files via Moodle by May 28th, 23:59.
- Selected groups will be asked to present their results in class on June 1st.
- At least one team member must be available to present in-person or via Zoom.

Data Classification Task: Grading

- Completing all tasks **is mandatory** to pass the course!
- The report is worth **37 points** and the slides **3 points**
- **Grading criteria** for the report in the task description document on Moodle
- No late submissions will be accepted. Make sure you upload your files in time. Check in with your teammate often, and get in touch with us at least two weeks before the deadline if they are unresponsive.

A note on Regrouping

- We are offering a one-time window for adjusting groups. You have until **Friday, April 24th** to request a change in your group.
- Send an email to mlpc-cp@jku.at with the following information:
 - ☐ Current Team Name
 - ☐ Current Team Mate
 - ☐ Reason for Regrouping Request
 - ☐ New Desired Team Mate (can be empty)
- Include your new teammate in the CC
- After this deadline, we merge the remaining one-person groups into groups of two
- **No further team changes will be allowed after this deadline.**

Data Classification Task: Summary

- Completion of Task 4 is mandatory
- Answer all **four aspects** and the **corresponding questions** in your **written report**
- Use the **L^AT_EX template**, stick to the **page limit** (6 pages, at most 2000 words)
- Create a **slide deck** which tackles the **one aspect assigned to your group**. 4 slides + 1 title slide max.
- Upload both until **May 28th** to get up to **40 points**
- Please note that de-registration is no longer possible. A certificate will be issued to all students who submitted the Data Exploration Task.